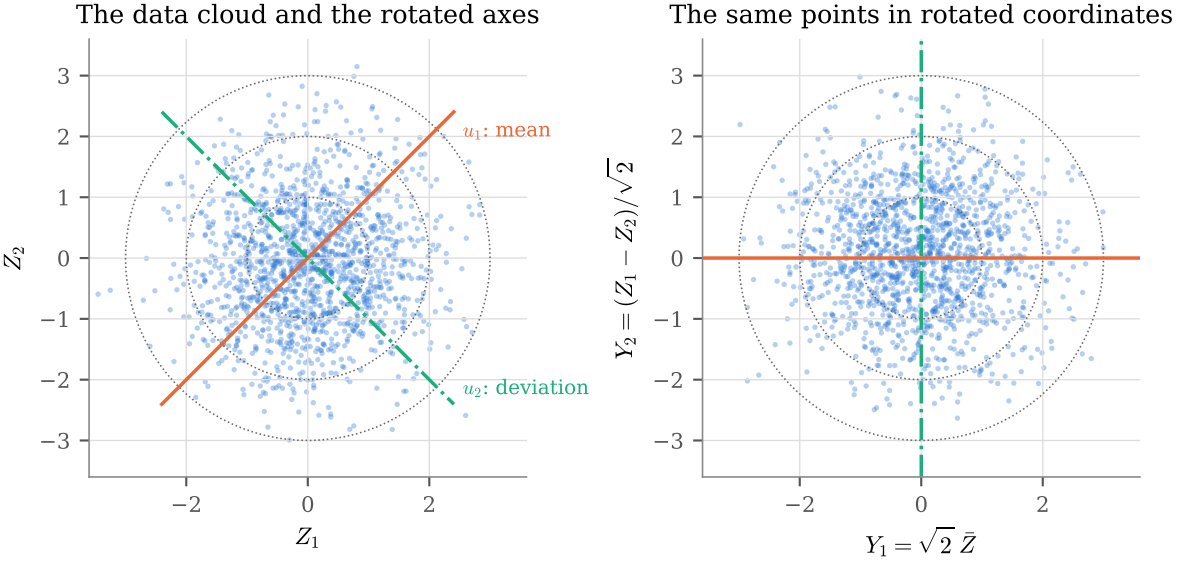

The data cloud and the rotated axes
u_1: mean
u_2: deviation
Z_1
Z_2
The same points in rotated coordinates
$Y_1 = \sqrt{2}\,\bar{Z}$
$Y_2 = (Z_1 - Z_2)/\sqrt{2}$